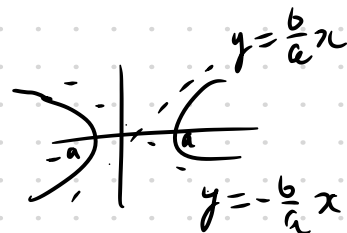


Special Cases Continued

Consider $(x, y) \in \mathbb{R}^2$: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



Theorem: Consider $(x, y) \in \mathbb{R}^2$ satisfying
 $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$

Then we have the following cases:

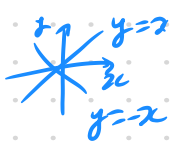
$B^2 - 4AC < 0 \Rightarrow$ Ellipse

$B^2 - 4AC = 0 \Rightarrow$ Parabola

$B^2 - 4AC > 0 \Rightarrow$ Hyperbola

← Use for proofs

← Warns for up to rotation and translation

$x^2 - y^2 = 0 \Rightarrow$  'degenerate case'
 "line pair"

Consider $y = mx + c$ and $y^2 - x^2 = 1$

Are there any values of m leading to tangency to the hyperbola

$$y^2 - x^2 = 1 \Rightarrow (mx + c)^2 - x^2 = 1$$

$$\Rightarrow m^2 x^2 - x^2 = 1$$

$$\Rightarrow (m^2 - 1)x^2 - 1 = 0$$

$$\Rightarrow m = \sqrt{\frac{1}{x^2} + 1}$$

$$-4(m^2 - 1)(-1) = 4m^2 - 4$$

$$\Rightarrow m^2 - 1 = 0 \Rightarrow \boxed{m = \pm 1}$$

$$y^2 - x^2 = 1$$

$$\rightarrow \frac{dy}{dx} - x = 0$$

$$\Rightarrow \frac{x}{y} = \frac{dy}{dx}$$

$$(m^2 + 1)x^2 + 1 = 0$$

Parametric equation

$x \equiv x(\theta)$ and $y \equiv y(\theta)$, where θ is the parameter, $\theta \in \mathbb{R}$

x, y are functions of θ